

E-Commerce Customer & Sales Analysis Using Python EDA

1. Project Title

E-Commerce Customer & Sales Analysis Using Exploratory Data Analysis (EDA) in Python

2. Dataset Description

- Rows: 11251 and Columns: 15
- Customers Detail (User_Id, Cust_name, Gender, Age Group, Age, Marital_Status, Occupation)
- Geography (State, Zone)
- Products Detail (Product_ID, Product_Category, Orders, Amount, Status, unnamed1)

3. Tools & Technologies

Programming Language

Python

Libraries

- **Pandas** – data manipulation and analysis
- **NumPy** – numerical calculations
- **Matplotlib** – data visualization
- **Seaborn** – statistical visualization
- **Jupyter Notebook** – development environment

4. Data Loading

- The dataset was imported into Python using the Pandas library.
- The dataset dimensions were checked using: `df.shape`
- Column names were examined using: `df.columns` convert columns name in proper case.
- The first few records were examined using: `df.head()`

5. Understanding the Dataset

- The structure of the dataset was examined using: `df.info()`
- Statistical information was obtained using: `df.describe()`

6. Data Cleaning

- Blank columns were deleted using `df.drop()`
- Checked for null values and handled according to the business meaning using `df.isnull()`
- Deleted unnecessary null values using `df.dropna()`.# drop null values

7. Data Consistency

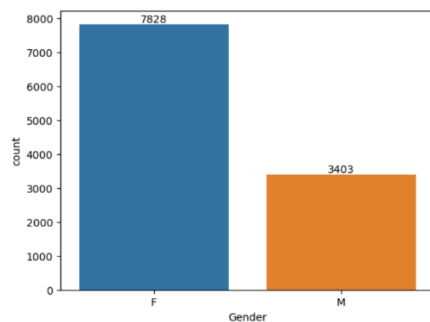
- Change data type as per requirement using `astype()`
- Added new column `M_Status_Name` to know marital status.

8. Duplicate Analysis

- Found duplicate records using `df.duplicated()` and removed them using `df.drop_duplicates()`

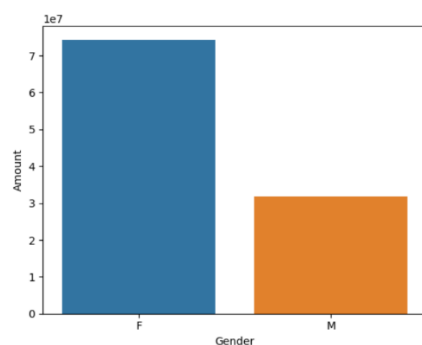
9. Important Business Questions Answered Through EDA

1. Which gender contributes the most number of sales?



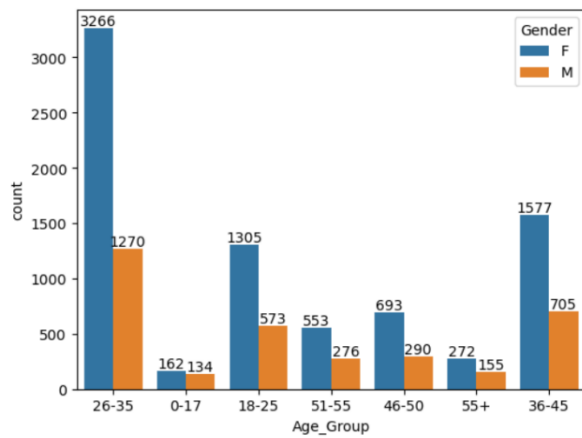
From the above graph we can see that females are purchasing more than men.

2. Which gender contributes the most revenue?



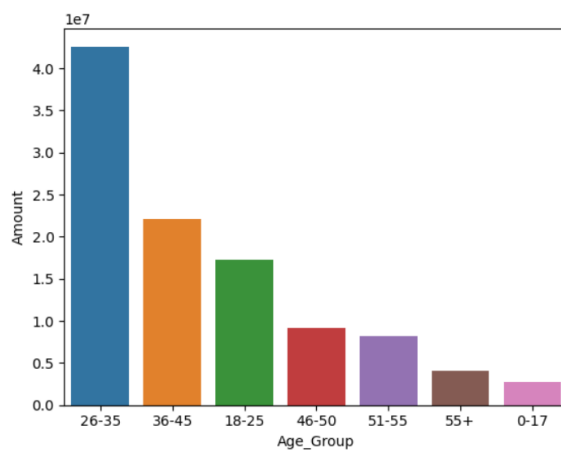
From the above graph we can see that the purchasing power of females are greater than men.

3. Which age group generates the highest sales?



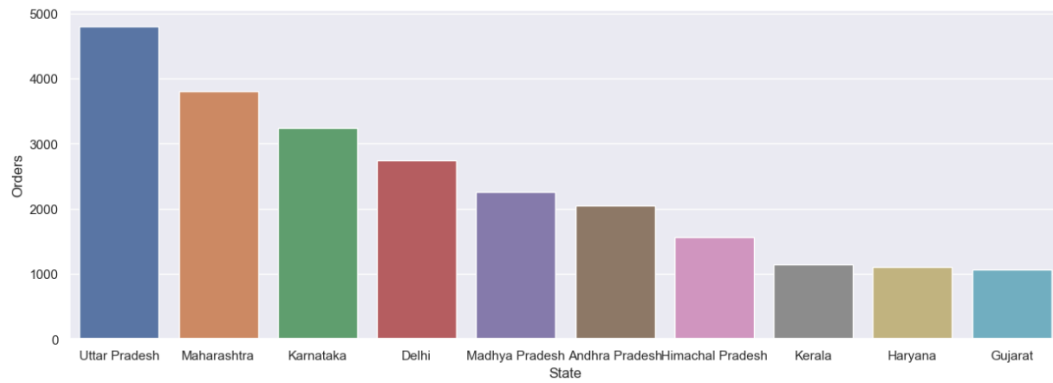
From the above graph we can see that highest sales are generated by age group between 26-35 years females.

4. Which age group generates the highest revenue?



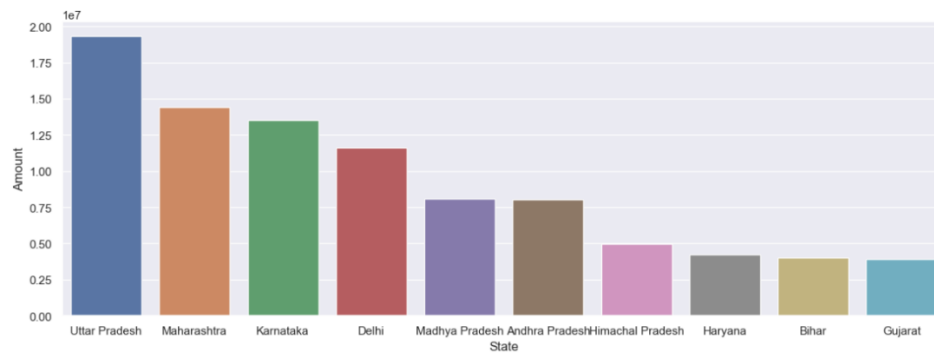
From the above graph we can see that highest revenue is generated by age group between 26-35 years female.

5. Which states have high order volumes?



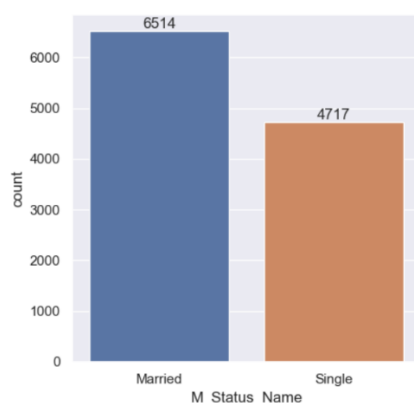
From the above graph we can see that most of the orders are from Uttar Pradesh, Maharashtra and Karnataka respectively.

6. Which states generate the highest revenue?



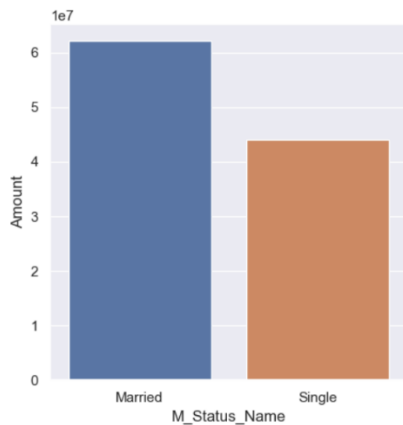
From the above graph we can see that most of the sales revenue came from Uttar Pradesh, Maharashtra and Karnataka respectively.

7. Which marital-status group spends more?



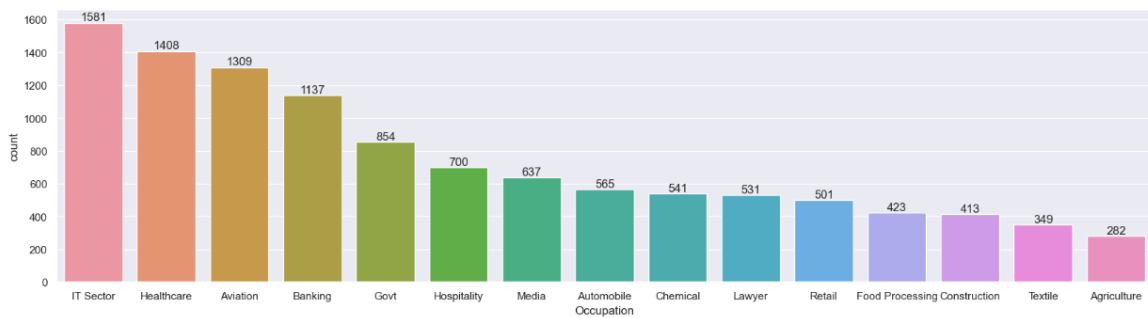
From the above graph we can see that most of the buyers are married (women).

8. Which marital-status group spends more?



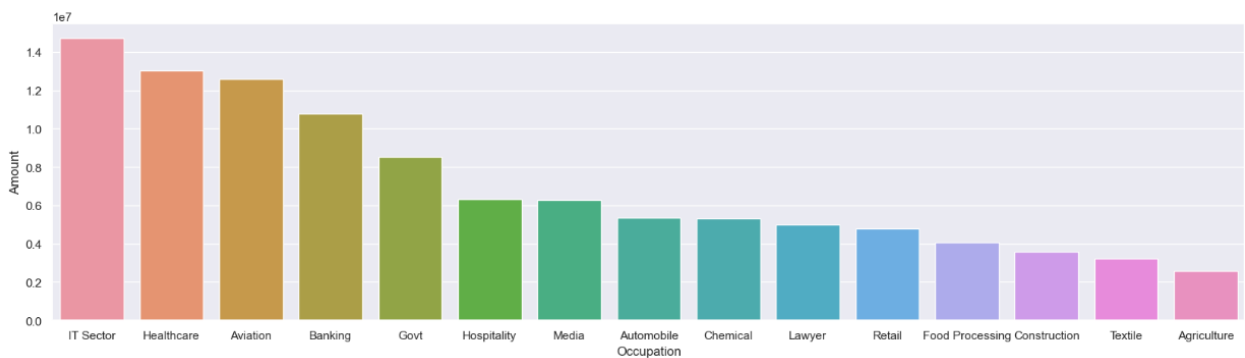
From the above graph we can see that most of the buyers are married (women) and they have high purchasing power.

9. Which occupations generate the highest sales?



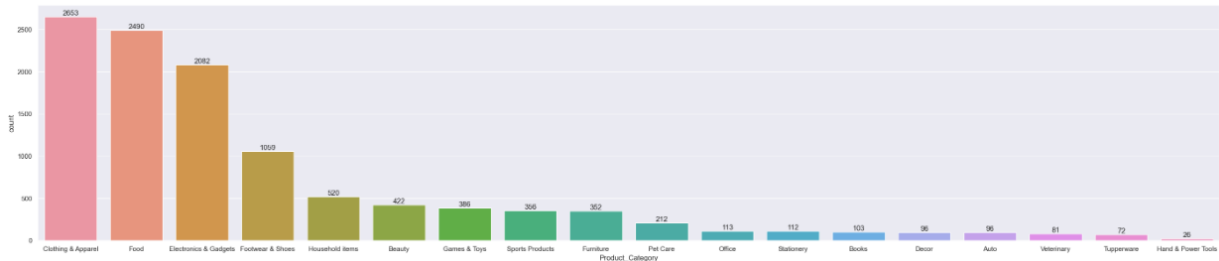
From the above graph we can see that most are the buyers are working in IT, Healthcare and Aviation Sector.

10. Which occupation has the highest revenue by number of orders?



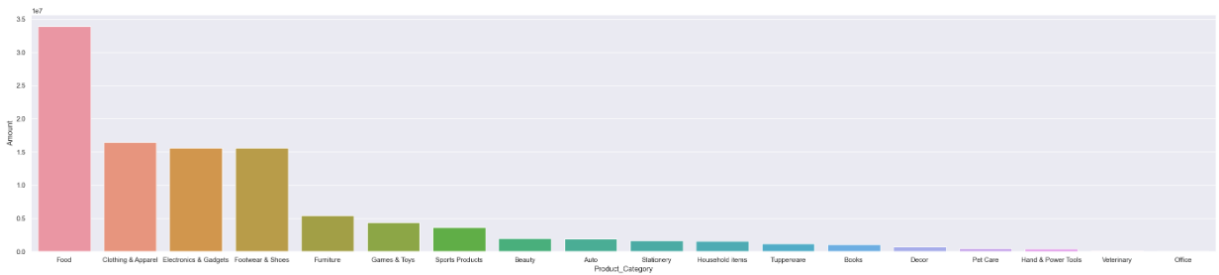
From the above graph we can see that most of the revenue is generated from the buyers working in IT, Healthcare and Aviation Sector.

11. Which product categories receive the highest number of orders?



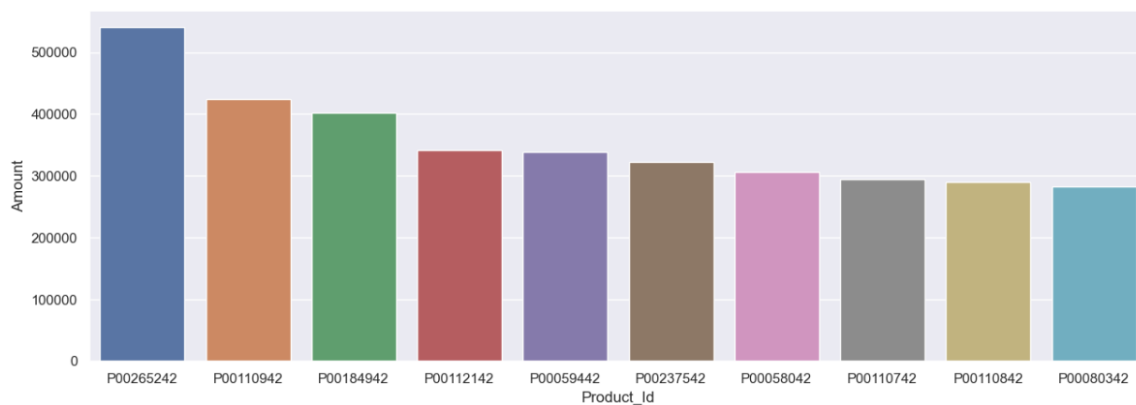
From the above graph we can see that most of the sold products are from Food, Clothing and Electronics.

12. Which product categories generate the highest revenue?



From above graphs we can see that most of the revenue is generated from Food, Clothing and Electronics.

13. Which top 10 product categories generate the highest revenue?



10. Conclusion

Married women age group of 26-35 years from UP, Maharashtra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from Food, Clothing and Electronics category.